power

File: power.kicad_sch

tube

File: tube.kicad_sch

mcu

File: mcu.kicad_sch

Sheets: power, tube, mcu. Inter-sheet nets are global labels.

Generated by tools/gen_schematics.py; values per docs/hardware.md

universal-nixie-display (open hardware)

Sheet: /

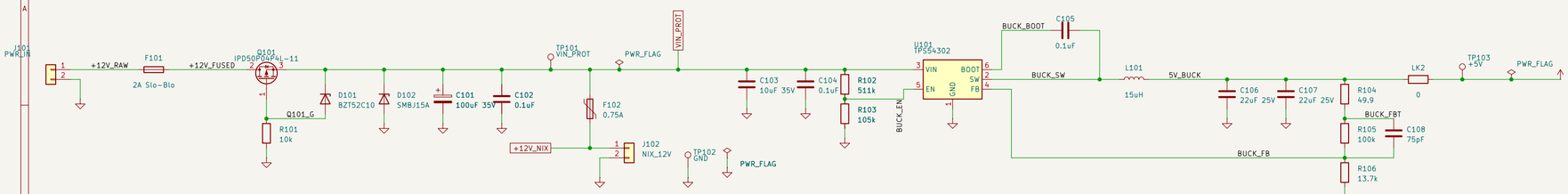
File: dekatron.kicad_sch

Title: Dekatron controller / PSU

Size: A4	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 1/4

12 V INPUT (D4)

Fuse, reverse-polarity P-MOSFET (gate clamped by D101), TVS, bulk; fused branch to the Nixie module

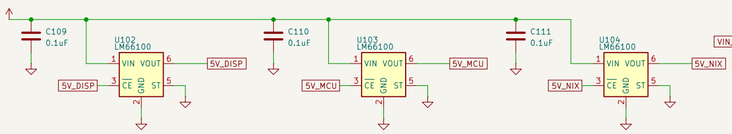


5 V BUCK

TPS54302, TI reference design (SLVSDG6C Fig. 7-1); 100k / 13.7k -> 4.95 V; UVLO 511k / 105k

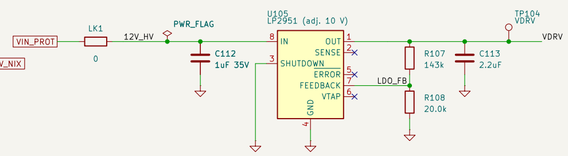
IDEAL DIODES (D6)

One LM66100 per external 5 V feed; CE tied to VOUT = always-on reverse-current blocking (datasheet SLVSEZ8A 8.3.2: CE to GND would not block); ST unused, to GND



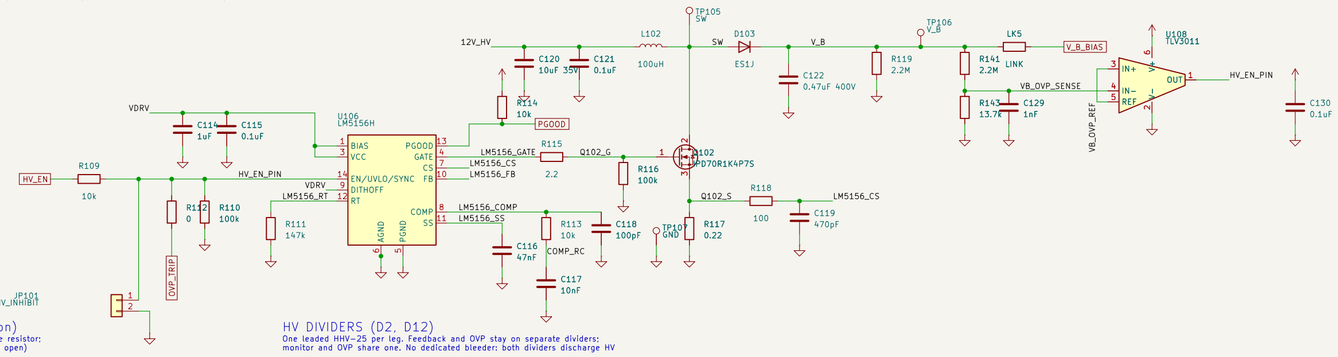
10 V GATE DRIVE

LP2951 feeds LM5156 BIAS = VCC (2.97-16 V mode): 143k / 20.0k -> 10.07 V



HV BOOST (D2, Phase 1B)

LM5156H, ~150 kHz DCM, 10 V gate drive, feedback from HV (tripler sheet area), CS filter against the pump surge: kit 100 uH inductor

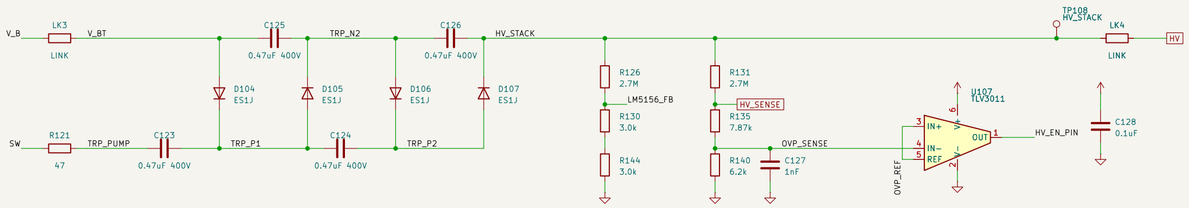


TRIPLER (series-stack Cockcroft-Walton)

Every capacitor and diode sees <= V_b; 47 ohm pump surge resistor; LK3 isolates the DC column for fault finding (never powered open)

HV DIVIDERS (D2, D12)

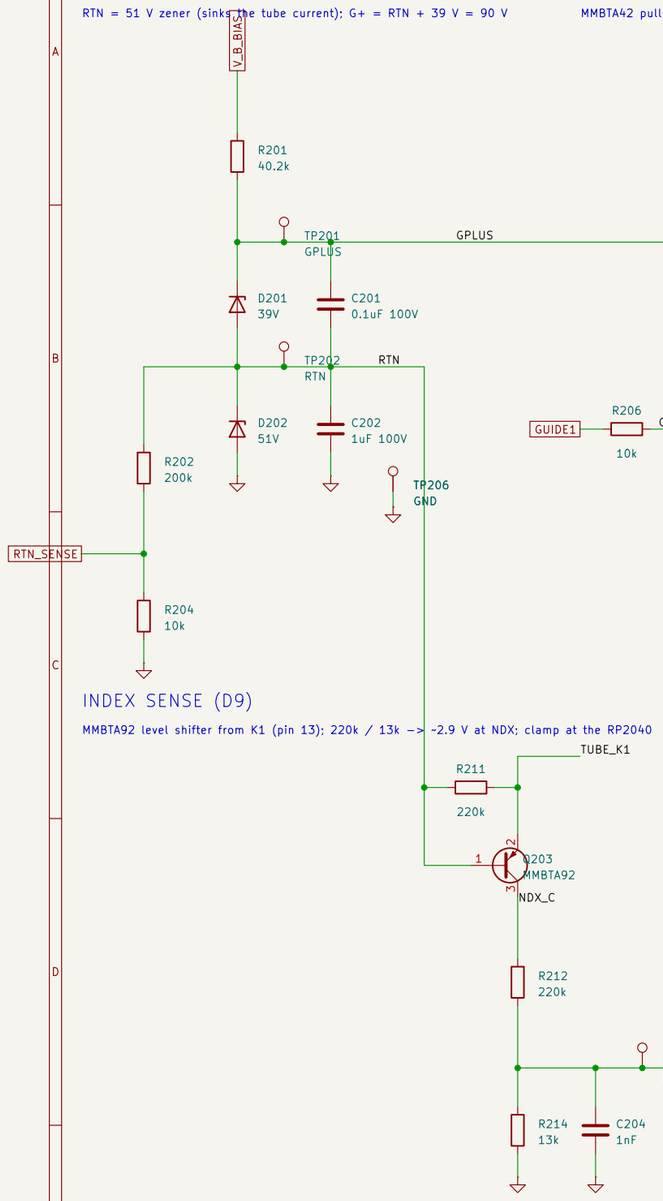
One leaded HHV-25 per leg. Feedback and OVP stay on separate dividers; monitor and OVP share one. No dedicated bleeder: both dividers discharge HV



Generated by tools/gen_schematics.py: values per docs/hardware.md
universal-nixie-display (open hardware)
Sheet: /power/
File: power.kicad_sch
Title: Power: 12 V input, 5 V, gate drive, HV converter
Size: A2 | Date: | **Rev: Rev A draft**
KiCad E.D.A. 10.0.5 | Id: 2/4

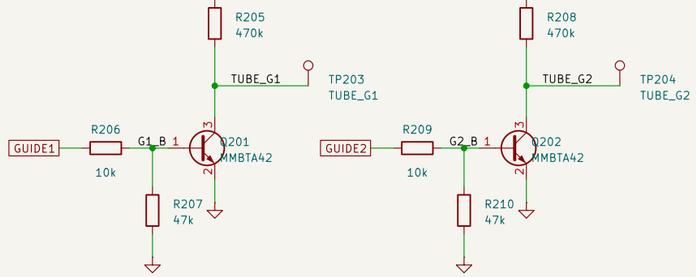
BIAS REGULATOR (D1, D8)

RTN = 51 V zener (sinks the tube current); G+ = RTN + 39 V = 90 V



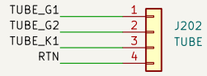
GUIDE DRIVERS (D8)

MMBTA42 pulls a guide from G+ to -0 V; 47k base pull-down = guides at rest in reset



TUBE HARNESS (D14)

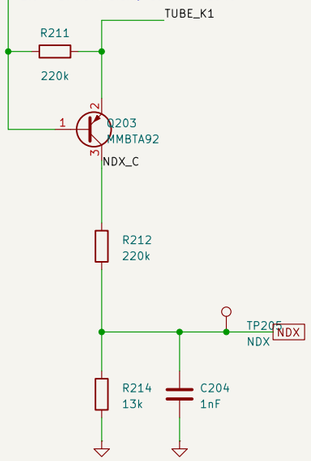
Wires soldered to board and socket; cathodes tied at the socket



A101 socket: anode 8, G1 5, G2 10, K1 13;
K0, K2–K9 (pins 1–4, 6, 7, 9, 11, 12) tied together onto RTN

INDEX SENSE (D9)

MMBTA92 level shifter from K1 (pin 13): 220k / 13k -> -2.9 V at NDX; clamp at the RP2040



ANODE STRING (D1)

2 x HHV-50 330k = 660k (330k + 470k = 800k for the 500 V setting); a shorted anode puts 0.23 W of 0.5 W on each 330k at 550 V; 67 mW per part at 0.45 mA



Generated by tools/gen_schematics.py; values per docs/hardware.md
universal-nixie-display (open hardware)

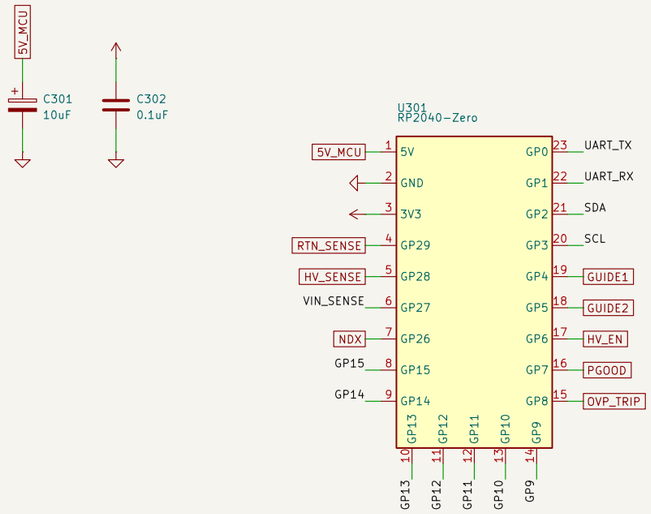
Sheet: /tube/
File: tube.kicad_sch

Title: Tube: G+ / RTN bias, guide drivers, index sense, anode, harness

Size: A3	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 3/4

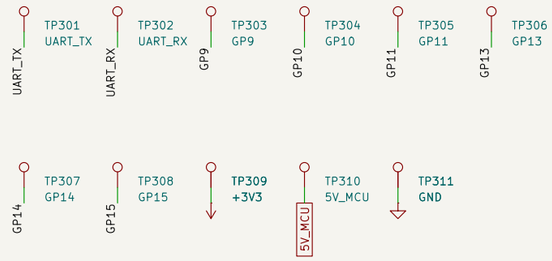
RP2040-Zero (pin map: docs/hardware.md)

Castellated, soldered by its edge pads; USB-C at the board edge



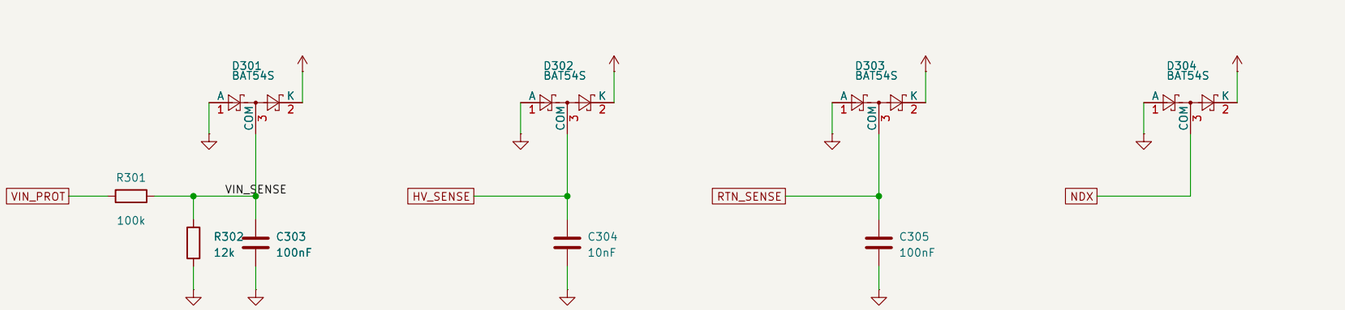
TEST PADS

Debug UART, spare GPIOs, supplies



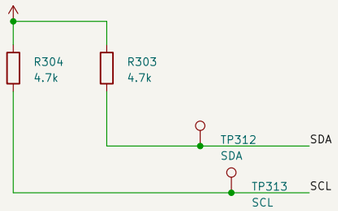
ADC INPUTS

BAT54S clamp to 3V3 / GND and a local reservoir capacitor at every input (NDX keeps its 1 nF on the tube sheet)



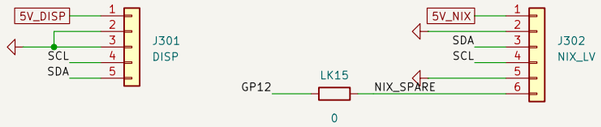
I2C (target, 100 kHz)

The bus pull-ups for the whole instrument live here (the display module has none)



HARNESS LANDINGS (D14, soldered)

DISP to the ESP32 display module, NIX_LV to the Nixie controller



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universal-nixie-display (open hardware)

Sheet: /mcu/		
File: mcu.kicad_sch		
Title: MCU: RP2040-Zero, ADC conditioning, I2C, harness landings		
Size: A3	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 4/4